

Model Solutions and Marking Notes

Regularized Regression — From Likelihood to Lasso · instructor copy

INSTRUCTOR COPY

Every number below comes from `scripts/analysis.py`, the single source of truth for the website, the notebooks and this document. Re-running it after a change to the data regenerates all three consistently. Student answers may differ in the last decimal depending on solver tolerances, but not in sign, magnitude or conclusion.

Task 1 · Maximum likelihood

1a · is_rain and traffic_speed on six trips

Correlation $r = -0.9360$, determinant of $X^T X = 4.4607$, condition number **30.2**. The estimate is $\hat{w} = (10.833, -34.224)$.

n	r	det $X^T X$	w · is_rain	w · traffic_speed	max move
6	-0.936	4.46	10.83	-34.22	—
7	-0.928	6.84	21.69	-23.85	10.85
8	-0.922	9.62	28.83	-16.77	7.15
9	-0.918	12.71	23.54	-23.84	7.08
10	-0.919	15.46	20.69	-27.38	3.53
11	-0.920	18.62	25.63	-18.60	8.78
12	-0.920	22.18	23.57	-20.50	2.06
13	-0.918	26.62	20.27	-24.20	3.70
14	-0.915	32.03	25.81	-21.58	5.54
15	-0.914	37.01	27.59	-19.32	2.26

MARKING NOTE

The largest one-row move is **₹10.85 per standard deviation**, and the coefficients are still oscillating at $n = 15$. Award marks for an explanation that ties the swing to the determinant and the $-r$ off-diagonal of the inverse. Observing only that the numbers move does not answer the question. Noting that the *sum* of the two coefficients is far more stable than either individually goes beyond what was asked and should be rewarded.

Prediction for trip 2 from the six-row fit: **₹146.03** against an actual charge of ₹150. Close, but that is luck: the model was fitted on six points, two of which were at the cap.

1b · is_rain and is_bad_weather on ten trips

On the first ten trips **every rainy trip also had traffic below 25 km/h**, so the two columns are identical. $X^T X$ is the all-ones matrix scaled by n , its determinant is exactly 0, and `np.linalg.inv`

raises `LinAlgError`. The problem first becomes solvable at $n = 14$, when a rainy trip with traffic above 25 km/h finally arrives.

THE ANSWER BEING LOOKED FOR IN STEP 3

The likelihood surface has a flat direction: adding c to one coefficient and subtracting c from the other changes no prediction and therefore no likelihood. There is no unique maximum, so "the MLE" is not well defined — a property of the model and data, not of the software. Ridge returns an answer because adding λI lifts the zero eigenvalue to $\lambda > 0$, breaking the tie in favour of the minimum-norm solution.

1c · Full eight-feature OLS

Condition number **37.6** — well conditioned. Train RMSE ₹9.15, test RMSE ₹13.24, test R^2 0.928.

Feature	Coefficient (₹ per s.d.)
is_peak	18.84
is_rain	12.65
traffic_speed	-16.52
drivers_avail	-0.98
is_event	7.42
is_airport	8.29
is_weekend	-0.17
open_requests	1.84

Task 2 · Ridge

α (sklearn scale)	w·is_peak	w·is_rain	gap	Train RMSE	Test RMSE	Max diff vs sklearn
0.1	18.79	12.61	6.18	9.15	13.23	7.1e-15
1	18.34	12.31	6.03	9.15	13.22	5.6e-15
10	15.61	10.53	5.08	9.40	13.29	8.0e-15
100	10.26	7.23	3.02	13.54	16.71	2.7e-15
1000	3.63	2.68	0.96	31.55	35.96	8.9e-16

The equal-split behaviour is visible in the `|gap|` column: the distance between the peak and rain coefficients falls monotonically as α grows. That is the answer to step 4 — there is no single α at which they "meet", they converge asymptotically, and a student who says so is more correct than one who reports a specific value.

5-fold CV on the training split chooses $\alpha = 1.75$ on the scikit-learn scale (= 0.01166 on the $1/2n$ -normalised scale), with CV RMSE ₹10.24 and test RMSE ₹13.21.

STEP 6 — THE PRIOR VERSUS THE EVIDENCE

The assumed prior gives $\lambda = 100$, which yields test RMSE ₹16.71 — **3.49 rupees worse** than the CV choice. The prior $\tau = 1$ asserts that coefficients are around ± 1 rupee per standard deviation; the data says they are around ± 10 . The prior is simply wrong, and cross-validation is detecting that. Full marks for saying the prior is mis-specified and that with 150 observations the likelihood should be allowed to dominate; partial marks for choosing CV without explaining why.

Task 3 · Lasso

3a · Agreement with scikit-learn

α	Sweeps to converge	Max difference	Non-zero (mine)	Non-zero (sklearn)
0.5	87	3.6e-07	7	7
2.0	87	3.9e-07	7	7
8.0	82	3.7e-07	6	6

Agreement to roughly $1e-7$, which is the coordinate-descent tolerance rather than a modelling difference. Students who report a mismatch have almost always used $\|y - Xw\|^2$ instead of $(1/2n)\|y - Xw\|^2$, a factor of $2n$.

Elimination order as α grows: `is_weekend` → `is_event` → `is_airport` → `is_rain` → `drivers_avail` → `open_requests` → `is_peak` → `traffic_speed`.

MARKING NOTE

`is_weekend` should die first — its true coefficient is exactly zero by construction. That `traffic_speed` and `is_peak` survive longest is the expected mirror image: they have the strongest marginal correlations with surge.

3b · New city, $p \gg n$

α	Method	Selected	True found	False pos.	Precision	Missed
1.0	lasso	26	6/6	20	0.23	—
1.0	enet90	51	6/6	45	0.12	—
2.0	lasso	18	6/6	12	0.33	—
2.0	enet90	38	6/6	32	0.16	—
4.0	lasso	10	6/6	4	0.60	—
4.0	enet90	27	6/6	21	0.22	—
6.0	lasso	7	6/6	1	0.86	—
6.0	enet90	23	6/6	17	0.26	—
8.0	lasso	7	6/6	1	0.86	—
8.0	enet90	22	6/6	16	0.27	—
10.0	lasso	7	6/6	1	0.86	—
10.0	enet90	20	6/6	14	0.30	—
12.0	lasso	6	5/6	1	0.83	<code>is_airport_pickup</code>
12.0	enet90	18	6/6	12	0.33	—
15.0	lasso	6	5/6	1	0.83	<code>is_airport_pickup</code>
15.0	enet90	15	6/6	9	0.40	—
20.0	lasso	2	2/6	0	1.00	<code>is_airport_pickup</code> , <code>is_event_nearby</code> , <code>is_peak</code> , <code>is_rain</code>

α	Method	Selected	True found	False pos.	Precision	Missed
20.0	enet90	8	4/6	4	0.50	is_airport_pickup, is_rain
25.0	lasso	1	1/6	0	1.00	is_airport_pickup, is_event_nearby, is_peak, is_rain, traffic_index
25.0	enet90	2	2/6	0	1.00	is_airport_pickup, is_event_nearby, is_peak, is_rain

Best Lasso operating point: $\alpha = 6$ — 7 features selected, all 6 real drivers recovered, 1 false positive, precision 0.86.

STEP 4 — THE ANSWER

At $\alpha = 12$ Lasso drops `is_airport_pickup`, a genuine driver, while retaining `feat_003`, which is pure noise correlated with it at $r = \mathbf{0.956}$. The l_1 penalty charges per non-zero coefficient and the two features carry nearly the same information, so paying for both is wasteful — but nothing in the objective prefers the *real* one. This is the selection-instability caveat made concrete, and a student who frames it as "Lasso made a mistake" has missed the point: the estimator did exactly what it was asked to do.

Step 5 — the counter-intuitive result. Elastic Net at $p = 0.5$ selects 63 features at $\alpha = 6$ and 27 features at $\alpha = 20$ — recall is perfect but precision collapses, because the l_2 term actively resists setting coefficients to zero. For *pure selection* in $p \gg n$, Lasso or a high p is the better tool. Award full marks for reporting this honestly; the grouping effect is a virtue only when the correlated partner is real, which here it is not.

Step 6: Lasso can select at most $n = 60$ features. The largest selection observed is well below that, so the bound never binds here.

Task 4 · Elastic Net and the grouping effect

`is_bad_weather` correlates with `is_rain` at **0.511** and with `traffic_speed` at **-0.644** — a genuine three-member group.

Step 3 · The three estimators at a common $\alpha = 2$

Estimator	w-is_rain	w-traffic_speed	w-is_bad_weather	Non-zero	Zeroed
Ridge	4.78	-7.72	3.41	9/9	—
Enet	5.72	-9.93	3.13	8/9	<code>is_weekend</code>
Lasso	9.60	-17.99	0.00	7/9	<code>is_weekend</code> , <code>is_bad_weather</code>

ALL THREE PREDICTIONS HOLD ON THIS DATA

Ridge keeps all nine features and spreads the weather signal across the three group members.

Lasso zeroes `is_bad_weather` and loads its weight onto `traffic_speed`, which nearly doubles.

Elastic Net keeps all three at moderate values while still zeroing `is_weekend`: the thing Ridge cannot do and Lasso does too aggressively.

Step 4 · ρ sweep

ρ	w-is_rain	w-traffic_speed	w-is_bad_weather	Non-zero
0.0	4.78	-7.72	3.41	9/9
0.1	4.93	-8.05	3.41	9/9
0.3	5.28	-8.85	3.35	8/9
0.5	5.72	-9.93	3.13	8/9
0.7	6.37	-11.63	2.50	8/9
0.9	7.71	-15.15	0.47	8/9
1.0	9.60	-17.99	0.00	7/9

Step 5 · Bootstrap selection frequency (200 resamples)

Feature	Selected by Lasso	Selected by Elastic Net
<code>is_peak</code>	100%	100%
<code>is_rain</code>	100%	100%
<code>traffic_speed</code>	100%	100%
<code>drivers_avail</code>	84%	100%
<code>is_event</code>	100%	100%
<code>is_airport</code>	100%	100%
<code>is_weekend</code>	3%	51%
<code>open_requests</code>	98%	100%
<code>is_bad_weather</code>	0%	100%

THIS TABLE IS THE TASK

`is_bad_weather` is selected by Lasso in **0%** of bootstrap resamples and by Elastic Net in **100%**: the grouping effect, measured rather than asserted. Note `is_weekend` too — Lasso keeps it 3% of the time, Elastic Net 51%. Elastic Net's reluctance to zero things is a cost as well as a benefit, and a student who points that out has read the table properly.

Task 5 · The surge engine

Quantity	Value
Demand / supply ratio	5.0
Raw model output	₹153.61
Surge after policy	₹150.00
Cap binding?	yes
Base fare	₹192.00
Final fare	₹342.00
Driver should reposition?	yes — surge exceeds the ₹50 threshold

STEP 3 — WHAT THE THRESHOLD ASSUMES

The "move if surge > ₹50" rule assumes the driver's cost of a 3 km deadhead — fuel, time, and the option value of staying where they are — is below ₹50, and that they will still be matched when they arrive. Neither is modelled. A good answer notices that the rule ignores the *probability* of getting the trip, which is precisely what falls as every other driver responds to the same signal.

Step 4 — supply sweep. Surge switches off at **17 drivers**, where the ratio first falls below 1.5. A rider requesting at 16 drivers pays the full capped surge; at 17 drivers they pay nothing extra. A ₹150 discontinuity produced by one driver moving one block is very hard to defend, and the best answers propose a smooth ramp — or note that the estimated ratio has its own uncertainty, which a hard threshold ignores entirely.

Step 5 — elasticity. The data contains no experimental price variation: prices were set *by a policy that responded to demand*, so price and demand are simultaneously determined and the regression coefficient on price would be hopelessly confounded. Only a randomised experiment — holding out a fraction of cell-buckets from the surge policy, or randomising the multiplier within a narrow band — can identify it. Full marks require naming the endogeneity, not merely saying "we need more data".

Bonus · The full posterior

MAP and Ridge agree to **1.2e-14** — machine precision. The posterior mode of a Gaussian is its mean, so this is an identity, not an approximation.

Feature	Mean	± 1.96 sd	95% interval	Covers zero?
is_peak	10.26	1.49	[8.76, 11.74]	no
is_rain	7.23	1.39	[5.84, 8.62]	no
traffic_speed	-12.02	1.52	[-13.54, -10.50]	no
drivers_avail	-6.12	1.49	[-7.61, -4.63]	no
is_event	3.63	1.28	[2.35, 4.92]	no
is_airport	4.94	1.24	[3.70, 6.18]	no
is_weekend	0.73	1.29	[-0.57, 2.02]	yes
open_requests	6.50	1.45	[5.05, 7.94]	no

1 of 8 intervals contain zero. Compare with Lasso, which set exactly that feature to zero at its CV-optimal α — the two methods agree about which feature carries no signal, but the posterior expresses it as a degree of belief rather than a hard deletion.

BKC predictive quote: **₹139.3 $\pm$ ₹10.2**, i.e. a 95% interval of [119, 159]. Of the total variance, **2.92 comes from uncertainty in w and 100 is irreducible noise** — so roughly 97% of it cannot be removed by collecting more data.

THE SENTENCE TO REFUSE (STEP 6)

The upper end of that interval exceeds the ₹150 cap, which means the honest quote would promise a price the policy layer forbids. Any sentence implying the rider might be charged above the cap should be rejected — as should any phrasing that presents the point estimate as a guarantee. A good answer also notices that showing a range invites the rider to wait and re-request, which changes their behaviour and therefore the demand the model was fitted on.

Deliverable 2 · The comparison table

Model	α	Train RMSE	Test RMSE	Test R^2	Non-zero	Dropped
OLS	—	9.08	12.89	0.931	9/9	—
RIDGE	0.0117	9.09	12.92	0.931	9/9	—
LASSO	0.1359	9.09	12.92	0.931	8/9	<code>is_weekend</code>
ENET	0.0215	9.09	12.92	0.931	9/9	—

THE POINT OF DELIVERABLE 3

Best and worst held-out RMSE differ by **₹0.03** on fares of a few hundred rupees. There is no statistically meaningful winner, and the strongest submissions say so and argue on operational grounds: Elastic Net for a short fare breakdown and stable explanations across refits; plain Ridge for one hyperparameter, a closed form and no feature silently vanishing after an overnight retrain. Penalise any answer that declares a winner on the third decimal of RMSE. Reward any that proposes re-running the comparison across several resampled splits first.